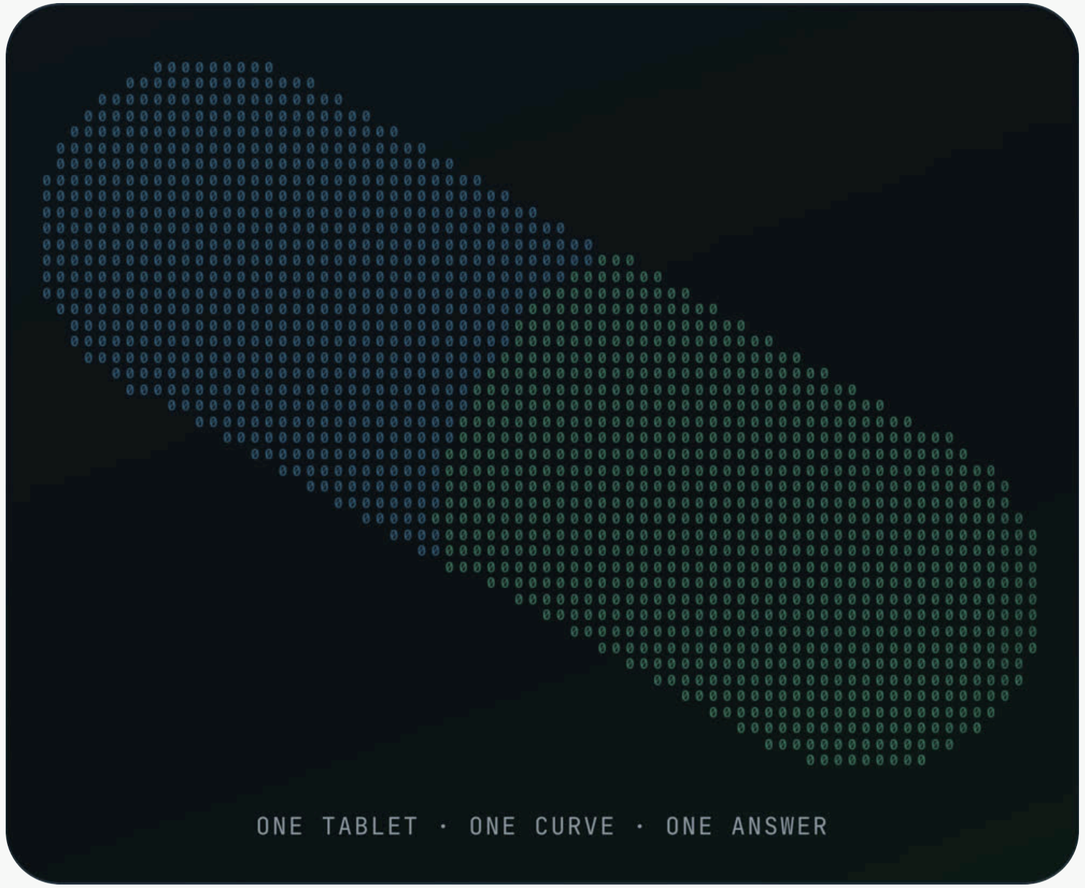
Pill

CAL HACKS 13.0 · IDEA TO DEMO · FIGMA

The box measures. *Pill explains.*

A $40 lab-in-a-box. One tablet, about three minutes, one of three answers. In a clinic, that answer decides whether the supplier gets paid.

PASS · NEEDS LAB · UNABLE TO VERIFY



Art · the site’s binary pill (pill.js), captured from pill-site.vercel.app

Same label. *Chalk inside.*

The bottle says vitamin B2. The blue line drops into the reference band, so a one-sensor tester would call it a pass. The green glow never moves.

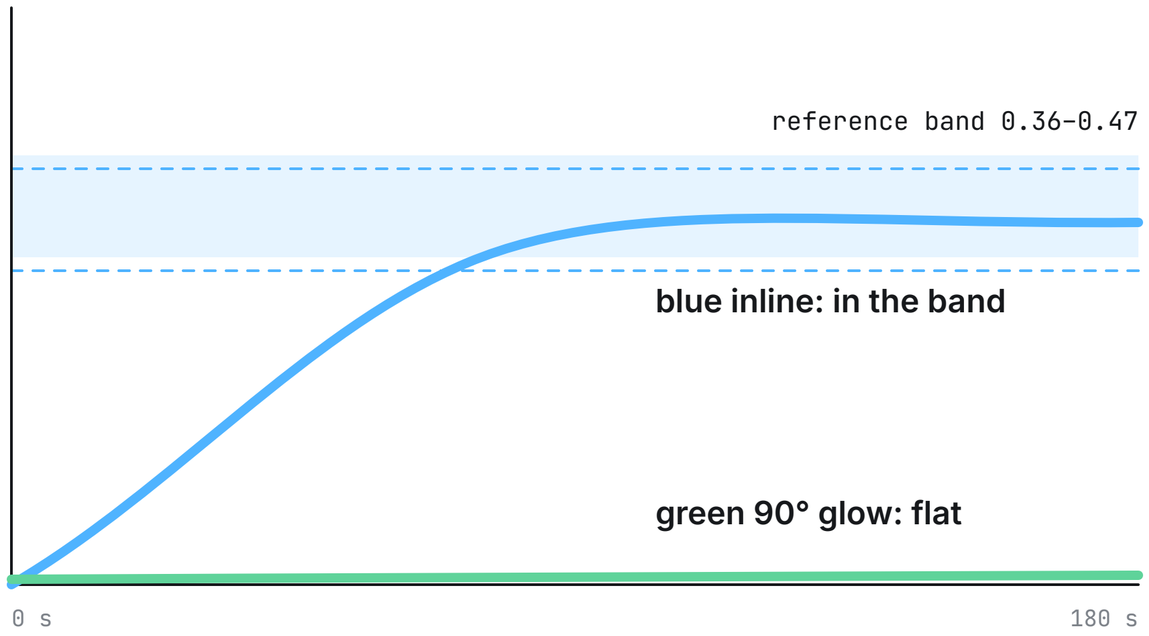


Illustration · the shape of replay TM-02, redrawn; not live sensor data.

RULE N2 · INLINE MOVED, GLOW DID NOT RISE

NEEDS LAB

A one-sensor tester would have passed it.

POLICY P2 · HOLD, DRAFT THE LAB REFERRAL

$480.00 held

SANDBOX · REPLAY

One in ten.

WHO estimates 1 in 10 medical products in low- and middle-income countries is substandard or falsified. The lab instrument that catches one costs $20,000 to $100,000.



Illustration · one tablet in ten. WHO, 2017: substandard and falsified products counted together.

116,000

extra malaria deaths a year in sub-Saharan Africa, modelled central estimate, from substandard and falsified antimalarials (range 64,000–158,000)

LSHTM model, cited by WHO, 2017

>300

children's deaths linked to contaminated cough syrups in three countries, 2022–23, most under five

WHO, Jan 2023

$30.5B

spent each year on substandard and falsified medical products, by WHO's estimate

WHO fact sheet, 2024

Lot numbers can be copied. Dissolution cannot.

A recall lookup trusts the label. Pill measures the tablet itself: one tablet in a cup of water inside a dark chamber, two light sensors watching it dissolve.

- **Arduino UNO Q · 3.3 V · 12-bit**
  Two photoresistors in 10 kΩ dividers from the 3V3 pin: A0 across the cup, A1 at 90° to the beam.
- **A dark reading every second**
  All LEDs off, both sensors read. Room light cancels out of the signal instead of leaking into it.
- **Blue, white, red in turn**
  Each LED lights alone and both sensors read again: light that gets through, light that scatters sideways.
- **One CSV line a second**
  ms, dark, blue, white, red for each sensor, at 115200 baud. The firmware decides nothing.

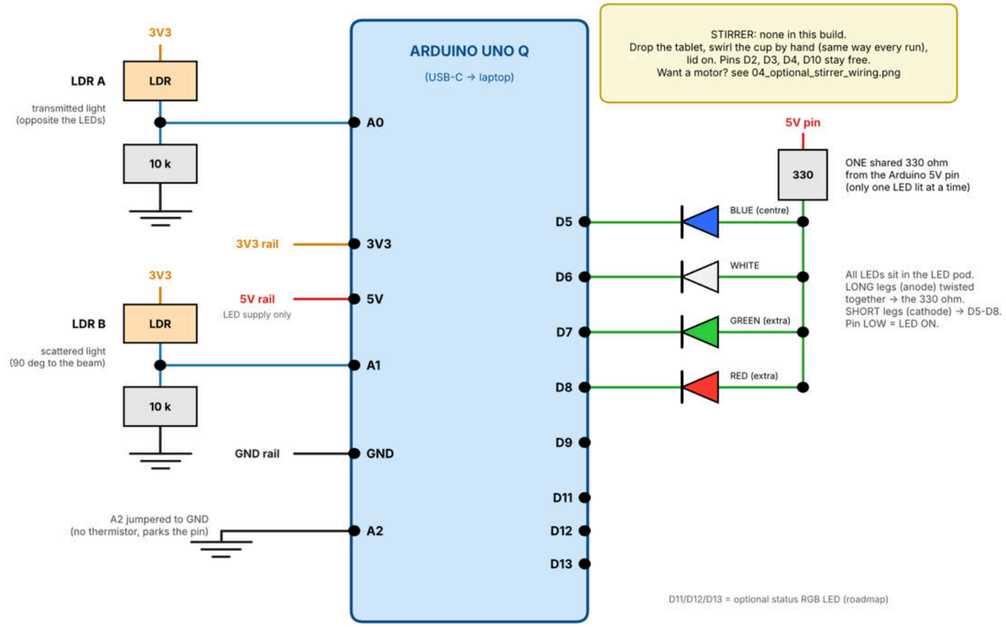


Fig 1 · UNO Q wiring (pillgate/diagrams): photoresistors on A0/A1 from 3V3; LEDs on D5–D8, active-low via one 330 Ω from 5V.

The station app.

Pill Gate runs on the laptop next to the box (Python, FastAPI). It guides the check and compares the run with a reference profile built from three or more runs of a known-good tablet.

01 — **Prepare**

`checks`

Lid closed, the LED reaches the sensor.

02 — **Clean water**

`10 s`

The blank must hold steady for 10 s.

03 — **Tablet in**

`lid opens`

The lid opening on both sensors marks the start.

04 — **Dissolving**

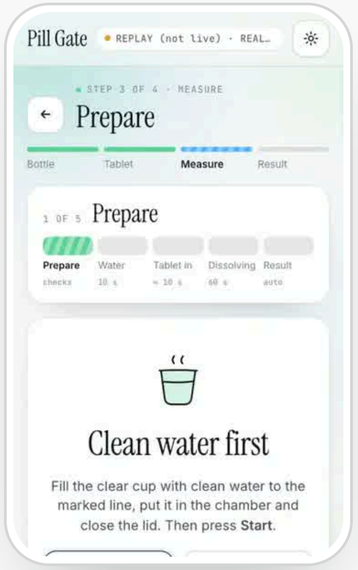
`60 s`

Absorbance and cloudiness curves, live.

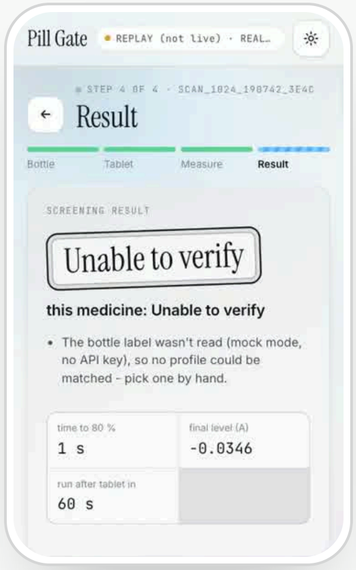
05 — **Result**

`auto`

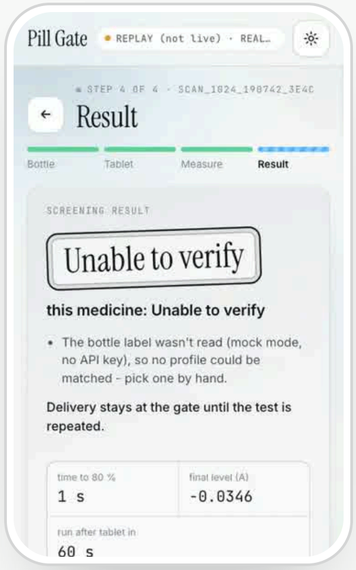
Compared with the reference profile.



Measure · the phase rail



Guided flow · step 6



Result · the answer

THREE ANSWERS, WRITTEN THIS WAY IN THE STATION APP

Passed the screen

Time-to-80 % and final colour inside the spread of the reference runs.

Needs lab testing

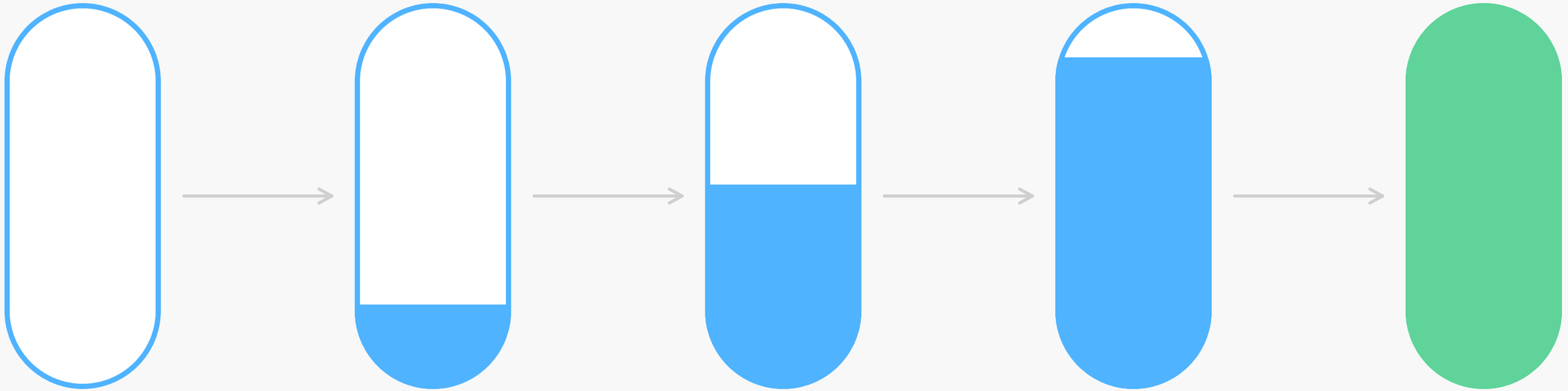
Outside it: slower, weaker, cloudier or clearer than the reference.

Unable to verify

No reference for this product, lid open too long, a noisy blank, or the curve never settled.

Three minutes, start to verdict.

The app draws the run as a pill filling to last night’s reference plateau (ten reference runs make the band). At 100% the curve has flattened and the seven rules run.



0%

`blank`

0:20 · dark, then blank. Lid open? No run.

22%

`dissolving`

0:40 · one tablet in. The paddle turns.

53%

`dissolving`

Blue light fades as the tablet dissolves.

86%

`settling`

The curve starts to flatten.

100%

`plateau`

3:00 · seven rules, in order. The rule that fired is printed.

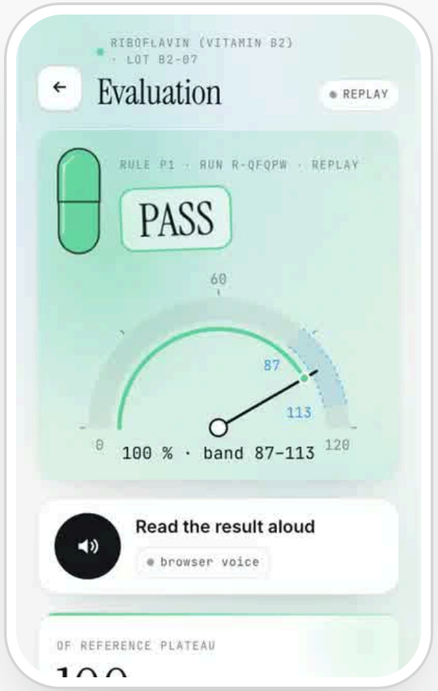
Illustration · example levels; the app draws them live from the sensor stream.

Three answers. *Never a guess.*

Seven rules, checked in order. A model writes the plain-English “why”; it never sets the verdict.

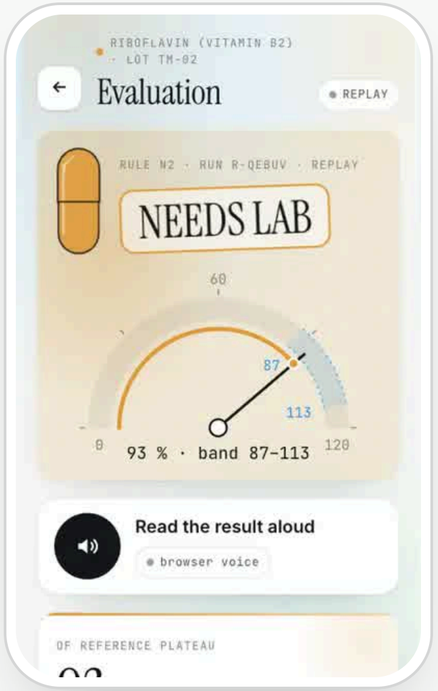
PASS

Plateau in the reference band and the glow rose with it (P1).



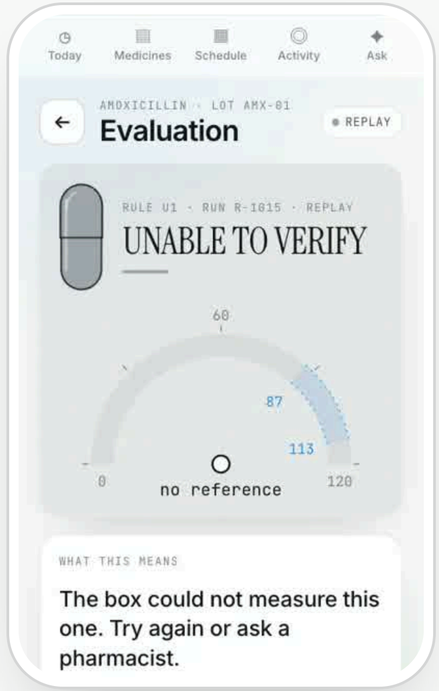
NEEDS LAB

Light dropped with no glow, no plateau, or a plateau outside the band (N1–N3).



UNABLE TO VERIFY

No reference for this drug, a fault, or too little data (U1–U3).



The verdict holds the payment.

SANDBOX · NESSIE TESTNET · BASE SEPOLIA

A clinic samples one tablet from each delivery. The policy agent reads the verdict, the recall check and the spend cap, and proposes. Only a pharmacist's tap releases money.

Clinic
wallet · USDC on Base
pays into escrow when the delivery is logged

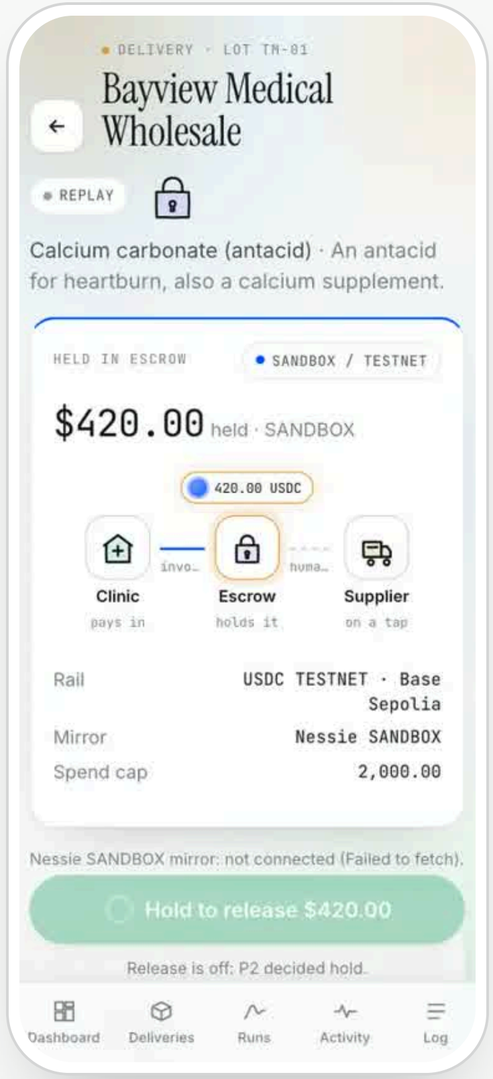
→

Escrow
Pill escrow
holds it until a PASS and a pharmacist's tap

→

Supplier
wallet · USDC on Base
paid only on release

- PASS → + the pharmacist's tap → escrow pays the supplier (P1)
- NEEDS LAB → held; the lab referral is drafted with the curve (P2)
- UNABLE TO VERIFY → escalated to a person; no money moves (P3)



app · delivery held under P2

The verdict is the invoice.

TESTNET · BASE SEPOLIA

0.05 USDC PER VERDICT

402

The server answers “Payment Required”: 0.05 USDC on Base Sepolia.

→

sign

The buyer’s wallet signs that exact payment; over its spend cap, it refuses.

→

settle

The facilitator settles it on Base Sepolia. TESTNET USDC only.

→

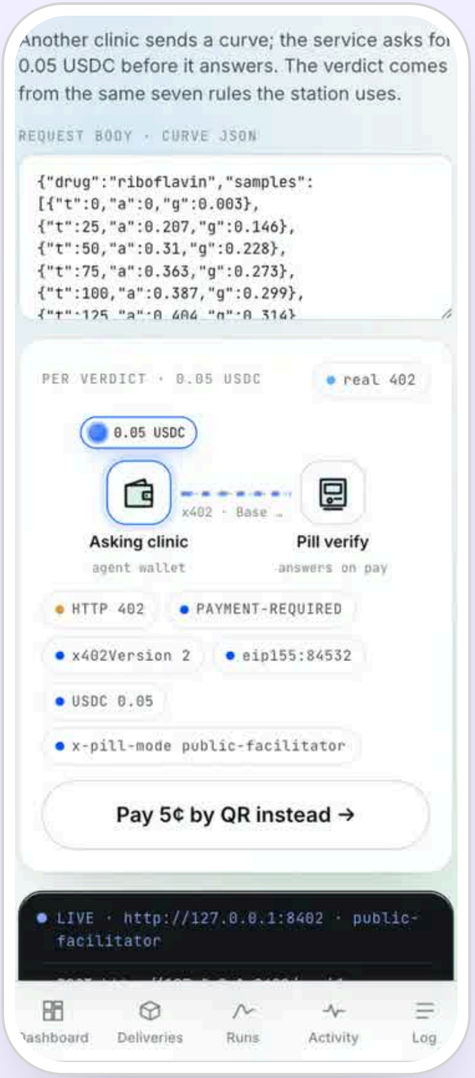
200

The server answers with the verdict and the rule id that fired.

01
Pay
Escrow → supplier in USDC, only after PASS and a pharmacist’s tap.

02
Buy
No local reference? Buy a verdict from another station over x402, capped per day.

03
Earn
Every station can sell /verify to other clinics.



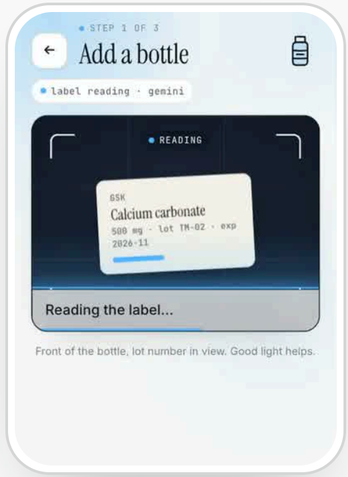
app · /verify over x402

One app: the touchscreen and the phone.

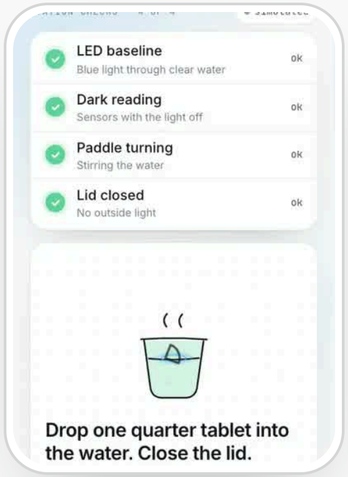
20 routes in React. All 303 app screenshots go into the Design file, one frame each, labelled with their route. These twelve are the newest pass.



welcome



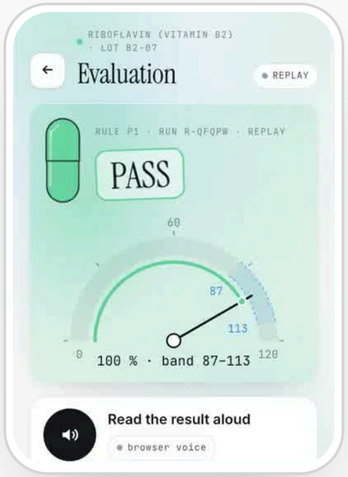
bottle



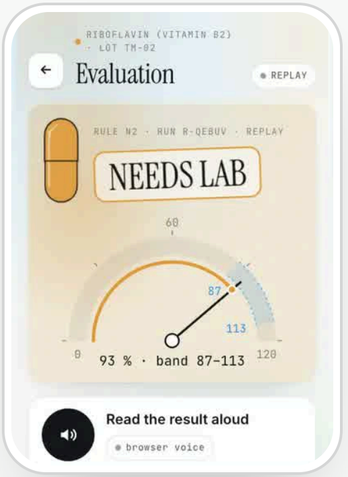
place



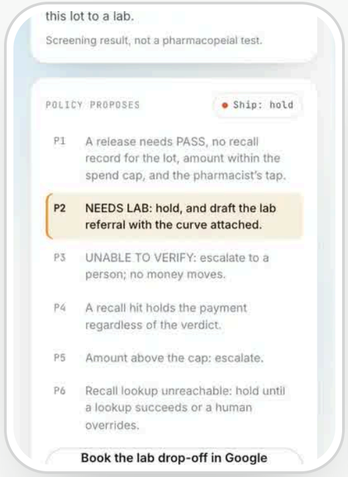
run



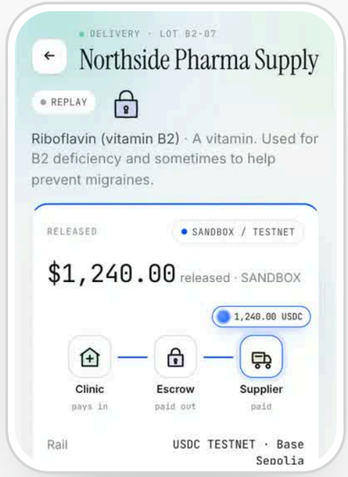
PASS



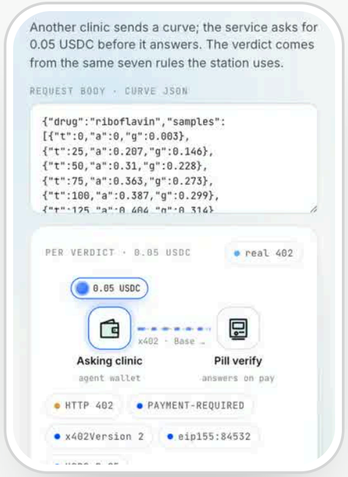
NEEDS LAB



held · P2



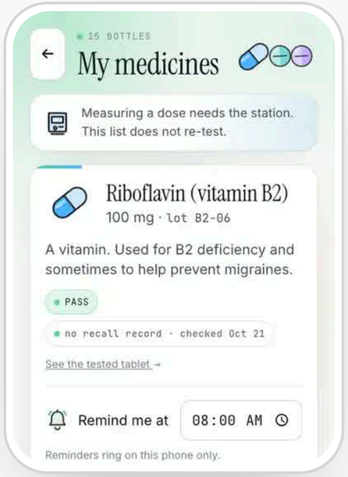
released



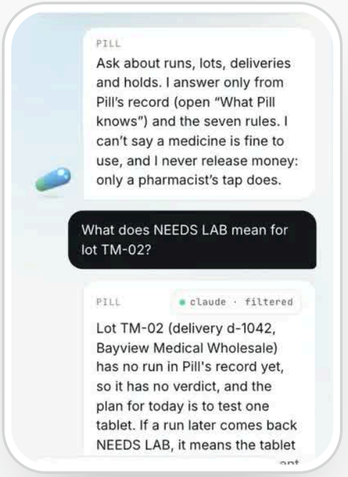
/verify · x402



home



medicines



ask

Six phases, one run.

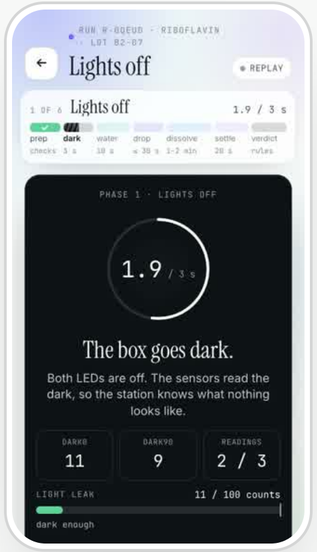
The app follows the station's own state machine. Each phase has its own time, its own look and its own check; a failed check stops the run and says why.

01

Lights off

3 s

LEDs off. The box must read dark; a light leak means the lid is open, and the run stops.

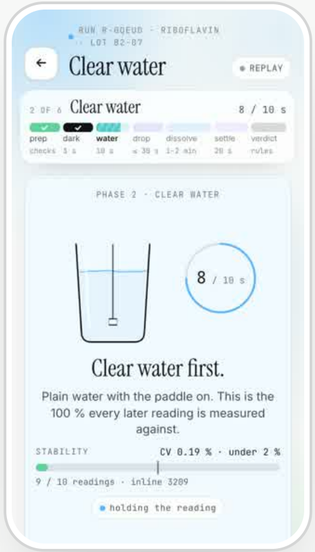


02

Clear water

10 s

Plain water, paddle on. The blank must hold steady (CV under 2 %) or it retries, up to 3 times.



03

Drop the tablet

≤ 30 s

It waits for the inline reading to fall 6 %. After 30 s the screen prompts.

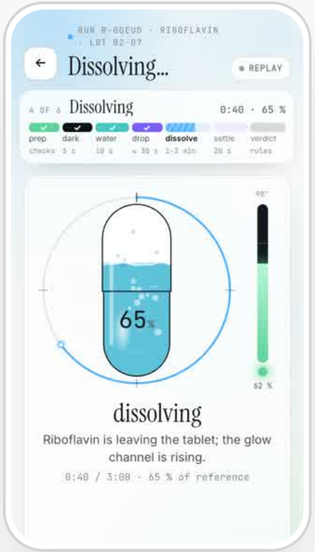


04

Dissolving

1–2 min

The pill fills, the curve falls, the glow rises. Glow flat while blue falls: the amber moment.

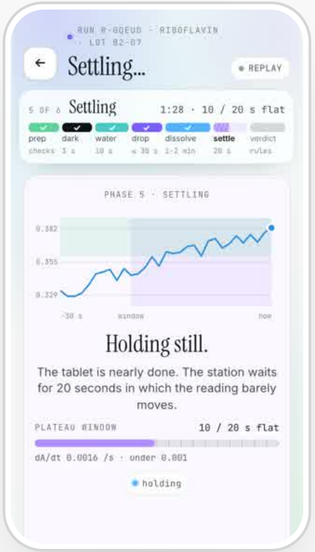


05

Settling

20 s window

Flat for 20 s (|dA/dt| < ε) means plateau. No plateau by 180 s: timeout.



06

Verdict

7 rules

Seven rules, in order. The rule that fired is printed with the word.

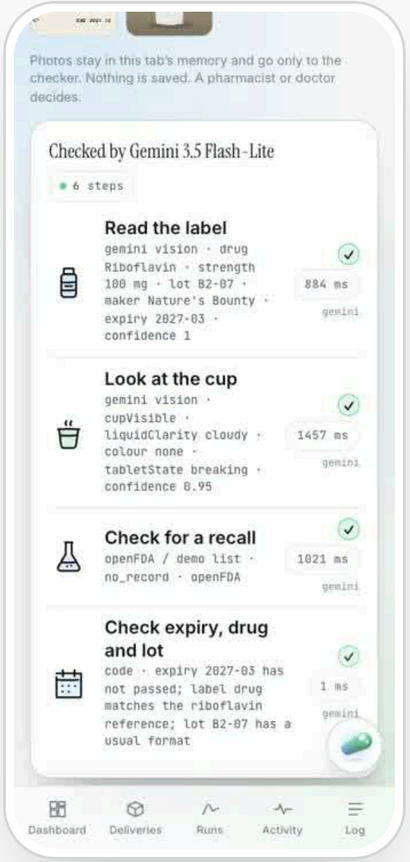


Gemini checks what the sensor says.

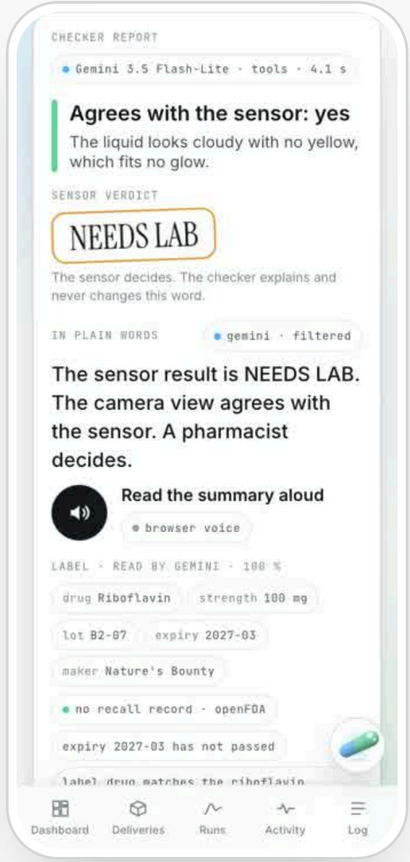
After the run, Gemini reads the label, looks at the cup and checks for a recall, then compares that with the sensor. Six steps, each logged with its own time.

1	**Read the label**	gemini vision
2	**Look at the cup**	gemini vision
3	**Check for a recall**	openFDA
4	**Check expiry, drug and lot**	code
5	**Compare with the sensor**	code
6	**Say it in plain words**	gemini · filtered

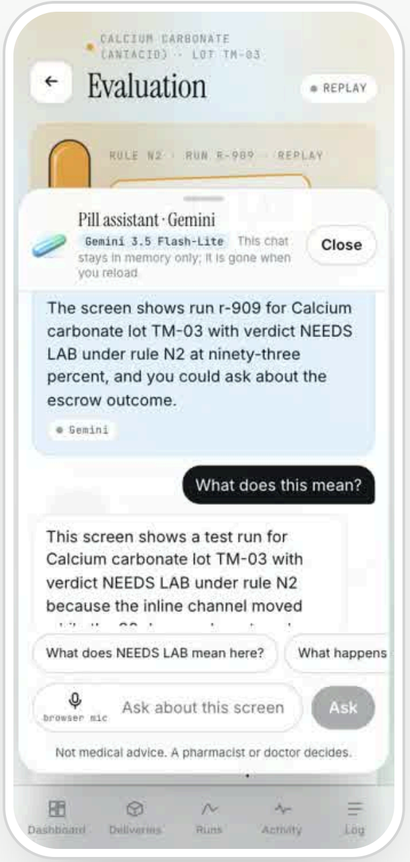
The sensor decides. The checker explains and never changes the word. A pharmacist or doctor decides.



Checker · the step log



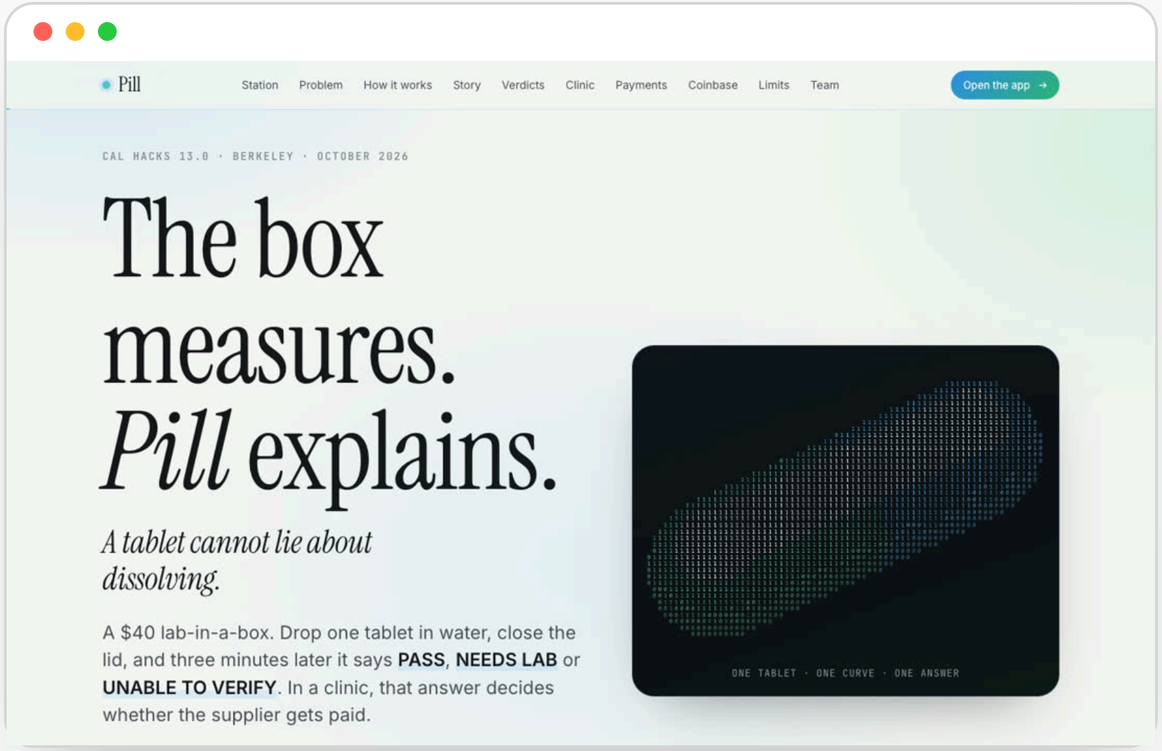
Checker · the report



Follower · on every screen

pill-site.vercel.app

Built code-first (Vite + anime.js) after a 16-persona panel scored a plain, honest page 8.0 against a copied storytelling site at 3.7. Words stay ink; colour lives in backgrounds and lines.



Hero · the binary pill, drawn in characters by pill.js

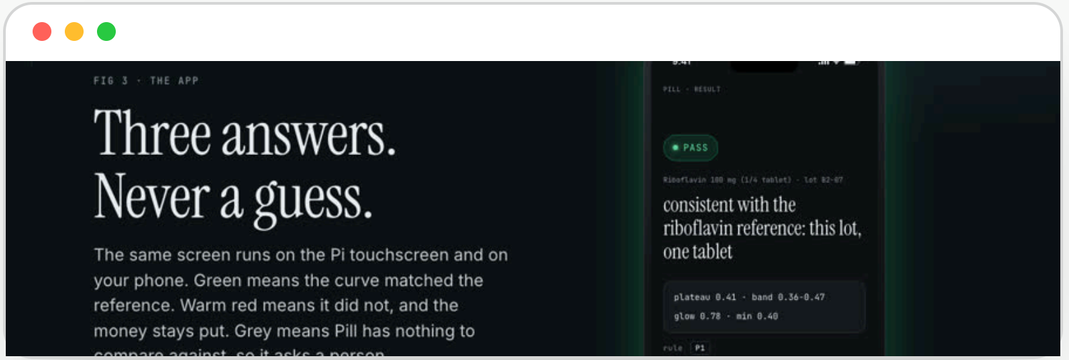
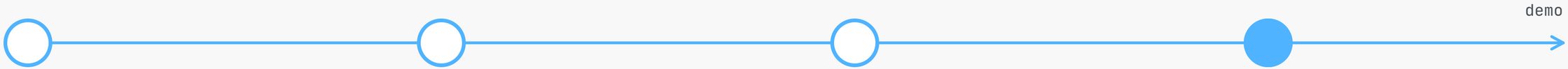


Fig 3 · three answers, the same screen as the app



Payments · the verdict is the invoice

Code first, then Figma from the code.



01
Ideation

CODE

- AI persona panels
- Web research

FIGMA

- FigJam board

≈255 min · 4/5

FIGMA STEP PENDING

02
Prototyping

CODE

- binary-pill.html
- Phone mock (Vite)

FIGMA

- Figma Make prompt

≈65 min · 3/5

FIGMA STEP PENDING

03
Handoff

CODE

- CSS :root, by hand

FIGMA

- Styles + variables
- Composio → CSS

≈7 min · 2/5

FIGMA STEP PENDING

04
Code generation

CODE

- Claude Code + 3 builders

FIGMA

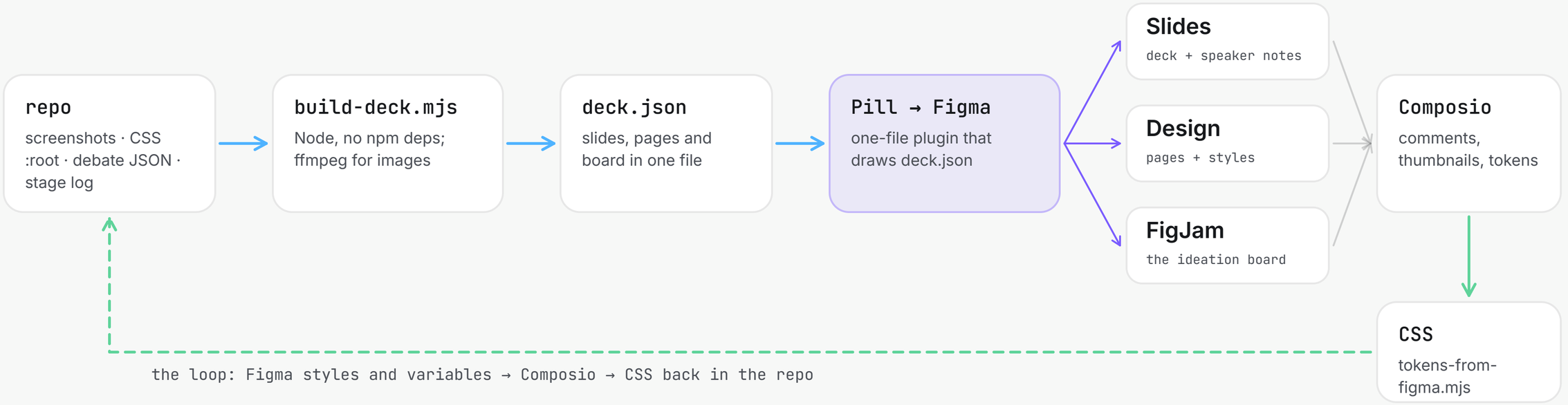
- Pill → Figma plugin

≈92 min · 4/5

FIGMA STEP PENDING

Honest order: no Figma was used Friday or Saturday morning. The Figma files are generated from the repo, and the log says so.

This deck was *generated from the repo.*



Recipe for code-first teams: keep the repo as the source of truth, emit one JSON, let a plugin draw it. One run gives designers editable layers, every screen, the tokens as styles, and a board of how you decided.

Why a plugin: the Figma REST API and our Composio connector can read, comment and set variables, but cannot create layers or slides. The Plugin API can.

Four stages, timed from git.

STAGE	TOOLS	MINUTES	RATING	WHAT BROKE	INTO THE PROJECT
Ideation	16 AI personas per panel, reference study; FigJam board generated after	≈255 min	4/5 PENDING	One model played all 16: agreement inflated (SD 0)	PLAN.md decisions, scope, site outline
Prototyping	binary-pill.html, phone mock in the site's JS; Figma Make prompt written	≈65 min	3/5 PENDING	Nobody could click it without a code editor	APP-PLAN's 12 screens, the pill fill
Handoff	CSS :root copied site → app; Figma styles + variables; Composio → CSS	≈7 min	2/5 PENDING	8 of 27 tokens arrived identical; no record of intent	App :root, Tokens page, tokens-from-figma.mjs
Code generation	Claude Code + 3 parallel builders; Pill → Figma plugin	≈92 min	4/5 PENDING	P6 hold rule bug; double deploy; REST API can't draw	App, payment service, this deck

PENDING = the Figma step is written but not yet run; its minutes and rating stay blank in the log, not estimated.

What broke, *and what it changed.*

The P6 hold rule

The scaffold held money only when the recall lookup was unreachable, so a delivery with no bottle attached could be proposed for release.

`Caught by a builder’s end-to-end test.`

A double deploy

A `tail -6` cut off Vercel’s output, the deploy looked unfinished and was re-run: two identical production deploys.

`Read the whole log before re-running.`

The REST API can’t draw

The Figma REST API and our Composio connector read, comment and set variables. Neither can create a layer or a slide.

`So the repo emits deck.json and a plugin draws it.`

8 of 27 tokens survived the hand copy

Site → app by hand: 8 identical, 7 changed value, 12 renamed or dropped. Nothing recorded which changes were meant.

`Tokens now go to Figma as variables and back as CSS.`

From hours to one click.

BEFORE · BY HAND · ESTIMATE

≈ 14.7 h

21 slides × 20 min + 421 screenshots × 1 min + 43 tokens × 1 min = 884 min

Per-unit minutes are our assumption, not a measurement.

AFTER · THE PLUGIN · MEASURED

38.1 s

repo → deck.json + deck.html, cold build on this machine (6.1 s this run). Plugin logic on a stub Figma API: 22 ms Slides · 274 ms Design · 4 ms FigJam.

Plugin run in real Figma: [fill after running]

bars to scale · after = cold build time

Every run places 421 screenshots (303 app, 118 site) and turns 43 CSS variables into Figma styles and variables, with no hand work. By hand, 8 of 27 site tokens reached the app unchanged.

Pill screens tablets. *It does not grade them.*

PASS MEANS

The dissolution curve of this one tablet, on this box, was consistent with the reference runs. No red flags found.

PASS DOES NOT MEAN

That the tablet is the right drug, the right dose, or free of contaminants. Anything Pill cannot compare goes to a lab, by design.

REAL VS SIMULATED, RIGHT NOW

Station hardware	REAL	Arduino UNO Q, 2 photoresistors, blue/white/red LEDs; one CSV line a second. One real aspirin run recorded Saturday. Reference profile not yet recorded.
Station app	REAL	Analysis, guided run, lid detection, guardrail; 14 tests pass. Live Claude scan ran Sat 19:19 on scan_1024_191752_f0c2 (replay of the real aspirin run): Unable to verify, no reference profile yet.
Clinic app curves	REPLAY	Synthetic curves (B2-07, TM-02, B2-LID, AMX-01), labelled REPLAY on screen; the week of history is fixture data.
Escrow · money	SIMULATED	Hold / release / refund simulated until a CDP key; the x402 402 and its signature check are real. SANDBOX / TESTNET only.

One build, many judges.

Figma · Idea to Demo
this deck, the plugin, the stage log

Hardware Hack
the station, on the table

Hyphenate · finance agents
policy agent: proposes, holds, escalates

Coinbase · x402 payment agents
/verify over x402, USDC escrow (TESTNET)

Capital One · Nessie
hold / release / refund (SANDBOX)

Microsoft · Built for the Real World
a clinic pharmacist's delivery check

Anthropic · Claude
the plain-English why; can only downgrade

People-First
personal mode, read aloud

Most Technically Complex
two-channel optics + rules + payments

MLH · Gemini API
reads the bottle label

MLH · ElevenLabs
reads the verdict aloud

MLH · Gen AI
models explain; rules decide

Next: *real runs, one clinic.*

1. **Calibration on the real box**
   Ten riboflavin reference runs and chalk runs, plotted on the site in place of the illustration.

2. **A second reference drug**
   Each new drug needs its own band. Until then it is UNABLE TO VERIFY, on purpose.

3. **One pharmacy, one week**
   A procurement officer samples one tablet per delivery; we log every decision and every override.

Ask: try to make it say PASS on the chalk. Then tell us which tablet to calibrate next, and introduce us to one clinic pharmacist.